SCHOOL OF COMPUTATION, INFORMATION AND TECHNOLOGY — INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Master's Thesis in Informatics

An Overlay for Social Networks based on Directed Acyclic Graphs on the Edge

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Ein Overlay für soziale Netzwerke auf der Grundlage gerichteter azyklischer Graphen auf der Kante

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I confirm that this master's thesis in informatics is my own work and I have documented all sources and material used.

Munich, 15.8.2026 Dan Bachar

Acknowledgments

Abstract

Recent crises, natural disasters, and large-scale protests have highlighted the significant reliance of people on online social networks and instant messaging platforms for communication. During crises, consolidated communication systems can become strained, partially or wholly destroyed, censored, or otherwise monitored. Permissionless distributed overlays are alternatives to consolidated communication realizations. System designers consider these overlays to be more resilient, since they distribute power and content control among many equal peers. These overlays introduce a massive performance overhead, as they rely on gossip and constant chatter between peers to disseminate the correct system state by reaching consensus, through many rounds of communication. This performance overheaf poses a risk to usablity of such systems in disaster zones, as peer devices often have limited connectivity and power supply, which together with the gossip protocol, can severely limit the peers' ability to communicate. In this paper, we propose a system architecture for unstructured peer-to-peer networks that achieves grassroots constitutional consensus based on efficient geometrical overlay design that trade achieve a higher throughput by using more edge computing. We use this system architecture to evaluate a decentralized social network deployed on smartphones using BLE connectivity, where users have total control of their data and privacy.

Kurzfassung

Recent crises, natural disasters, and large-scale protests have highlighted the significant reliance of people on online social networks and instant messaging platforms for communication. During crises, consolidated communication systems can become strained, partially or wholly destroyed, censored, or otherwise monitored. Permissionless distributed overlays are alternatives to consolidated communication realizations. System designers consider these overlays to be more resilient, since they distribute power and content control among many equal peers. These overlays introduce a massive performance overhead, as they rely on gossip and constant chatter between peers to disseminate the correct system state by reaching consensus, through many rounds of communication. This performance overheaf poses a risk to usablity of such systems in disaster zones, as peer devices often have limited connectivity and power supply, which together with the gossip protocol, can severely limit the peers' ability to communicate. In this paper, we propose a system architecture for unstructured peer-to-peer networks that achieves grassroots constitutional consensus based on efficient geometrical overlay design that trade achieve a higher throughput by using more edge computing. We use this system architecture to evaluate a decentralized social network deployed on smartphones using BLE connectivity, where users have total control of their data and privacy.

Contents

1. State-of-the-Art

1.1. Peer-to-peer Communication Technologies

Wifi Direct, Wifi Adhoc, Bluetooth BDR, Bluetooth Low Energy, Power usage of BLE vs. Direct, NAT, Hole-punching

1.2. Opportunistic Networking Systems

Twimight, Emergency Broadcast, COVID contact tracing

1.3. Routing in Opportunistic Networks

Flooding, Spray-and-Pray, PRoPHET

1.4. Decentralized Social Networks

Bluesky, Mastodon, Briar, Nostr, SBB, Bitchat

1.5. Grassroots Social Networks

Shapiro

2. Problem Analysis

2.1. Assumptions

- Disaster struck zone: Partially down communications, BLE first, mesh
- Politically oppresive regimes: various trust levels, edge rendezvous points, p2p
- Online monitoring: encryption through Noise handshake for key exchange
- More?

2.2. Design Goals

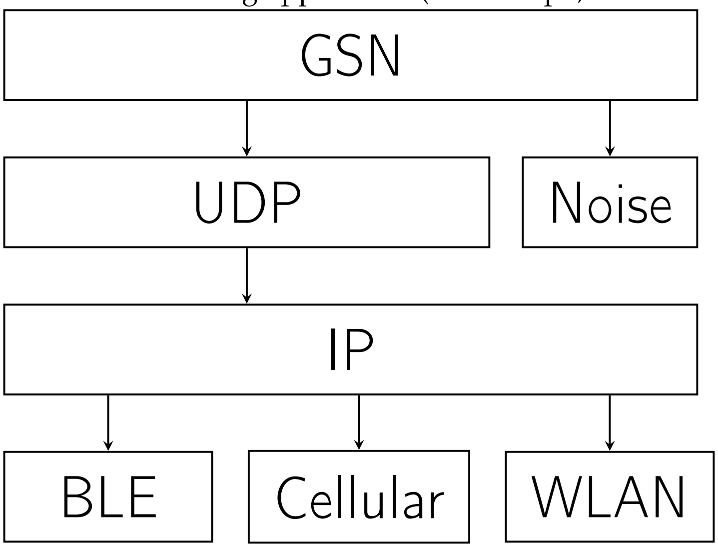
- Prefer local over remote communication
- When available, use the Internet-scale relation to relay messages over long distances
- Reduce attack interface for surveillance and monitoring as much as possible: expose the minimum amount of required information to facilitate communication
- Store-carry-forward: cache packets for unavailable recipients until a possible delivery attempt can be made
- Do not ignore existing mobile infrastructure to facilitate Internet-scale communication: edge rendezvous points and hole-punching
- Because the application may be used in various circumstances, let the user decide on various security-critical settings (i.e. use cloud RV point, friends, introduction facilitation)

3. Architecture

A demo chat app deployed on phones, which serves as the foundation to the blocklace-based future grassroots social network: argue why exactly these functions are needed.

3.1. System Architecture

The stack is organised into four layers. The lower two (transport + protocol) constitute the reusable library (‘lib/src/‘); the coordinator sits above them as the public façade; the topmost layer is the consuming application (in this repo, a demonstrator chat app under ‘lib/*.dart‘).



- Ed25519 identity
- Two transport protocols, one interface
- Message types: announce, message, friendship
- Friend vs. Stranger
- Trust Levels: cold calls
- Facilitation opt-in: edge RV points
- Always-on cloud RV points: optionally use RVs to facilitate connections
- Signaling over BT or mutual friends: hole-punching

- Simple messaging: text and media
- Blocks: ACK, NACK

4. Testbed

To evaluate the system architecture, we deployed it as a mobile application on a limited testbed of Android and iOS devices. During the deployment, usage statistics were recorded and uploaded to a secure TUM server on an opt-in basis.

5. Evaluation

6. Discussion

7. Conclusion

A. General Addenda

If there are several additions you want to add, but they do not fit into the thesis itself, they belong here.

A.1. Detailed Addition

Even sections are possible, but usually only used for several elements in, e.g. tables, images, etc.

B. Figures

B.1. Example 1

✓

B.2. Example 2

✗

List of Figures

List of Tables

bibliography.bib